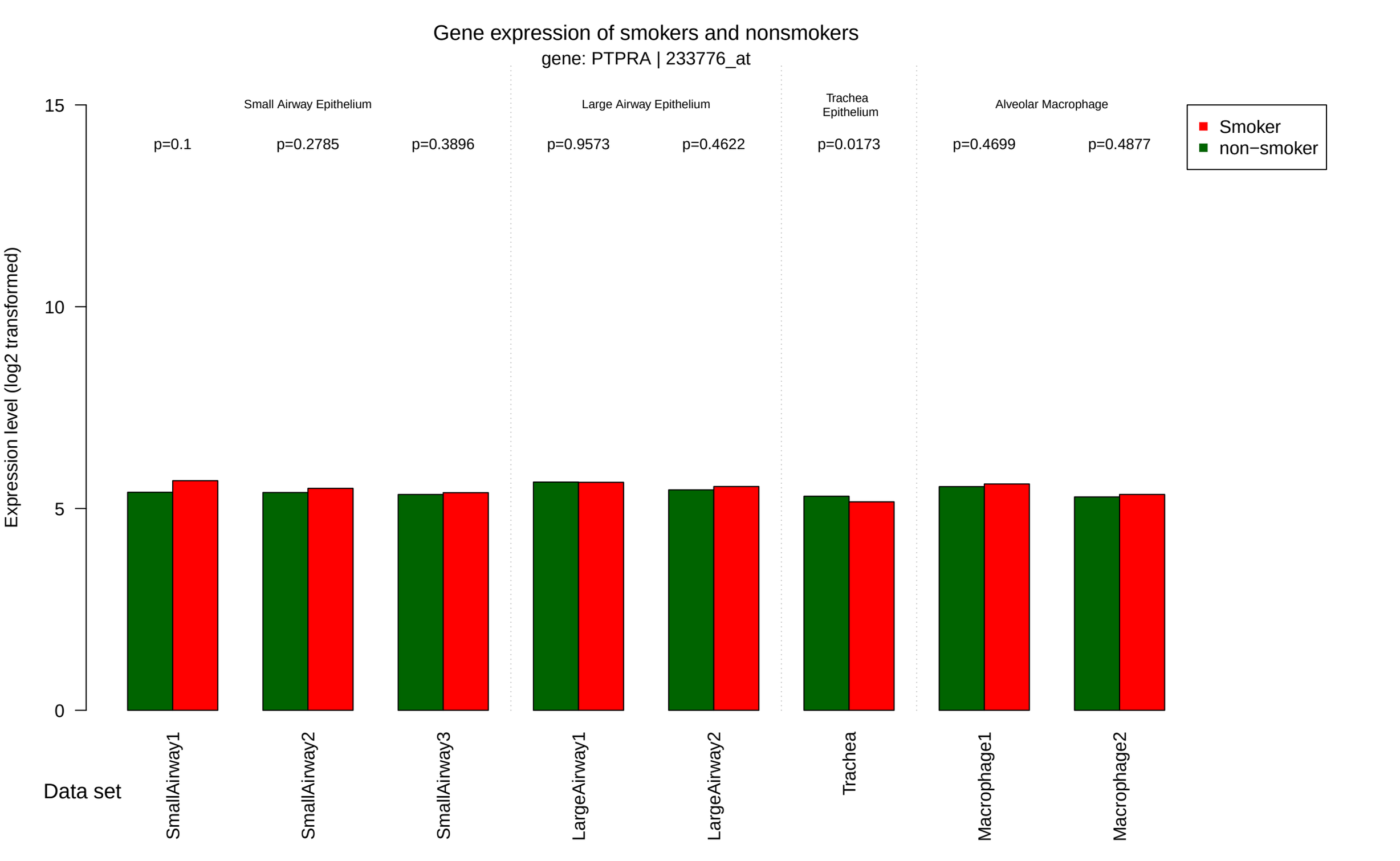
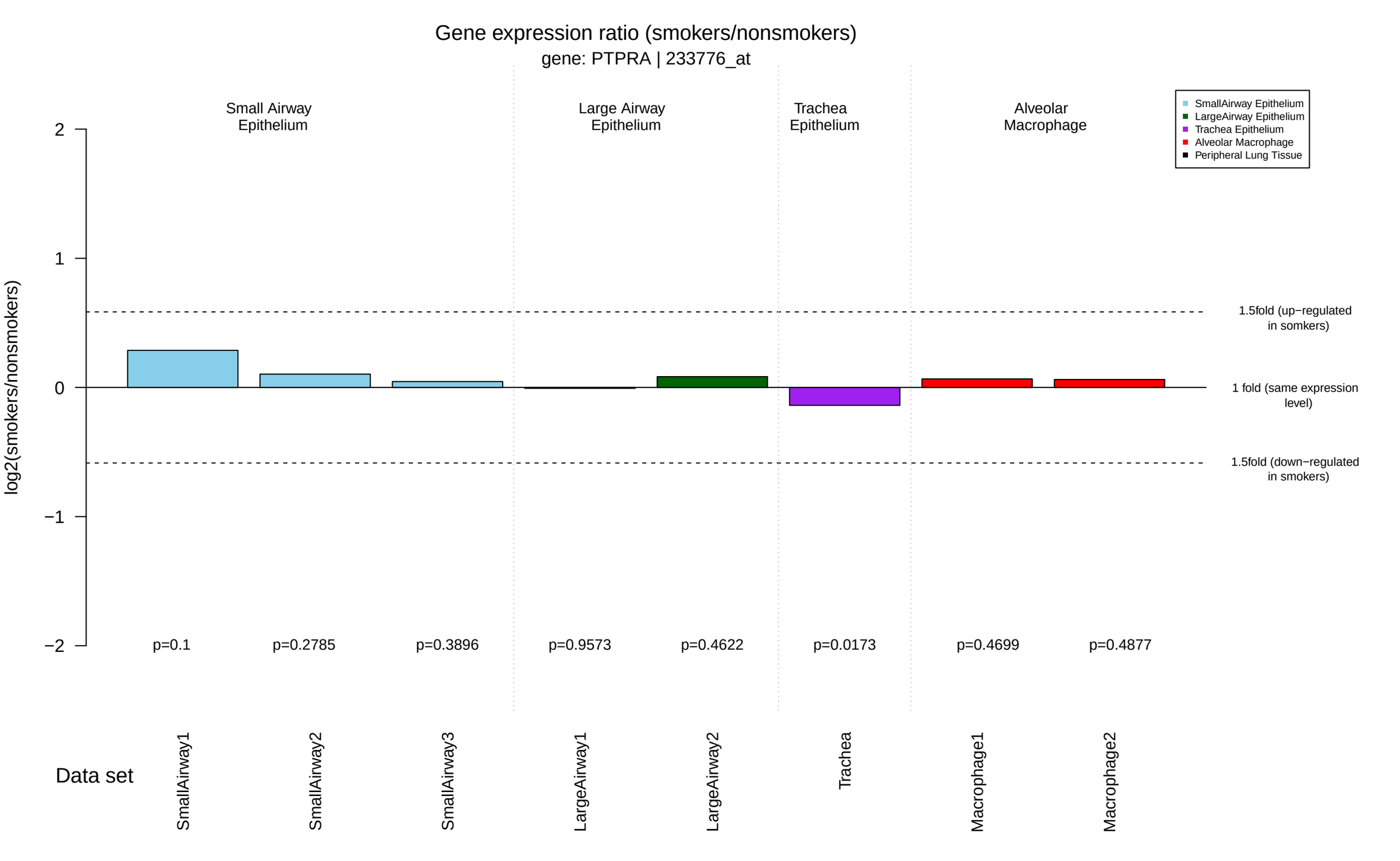
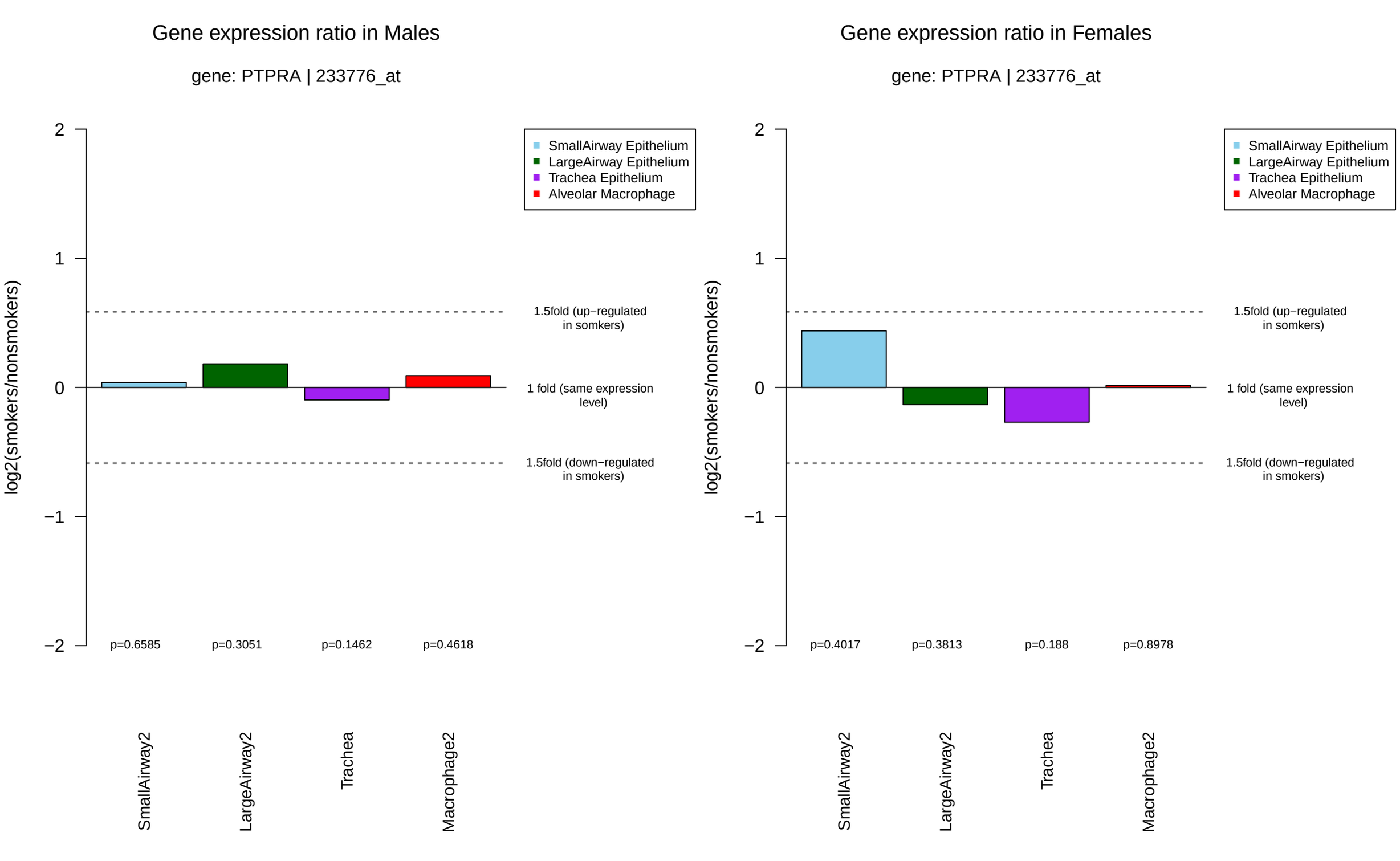
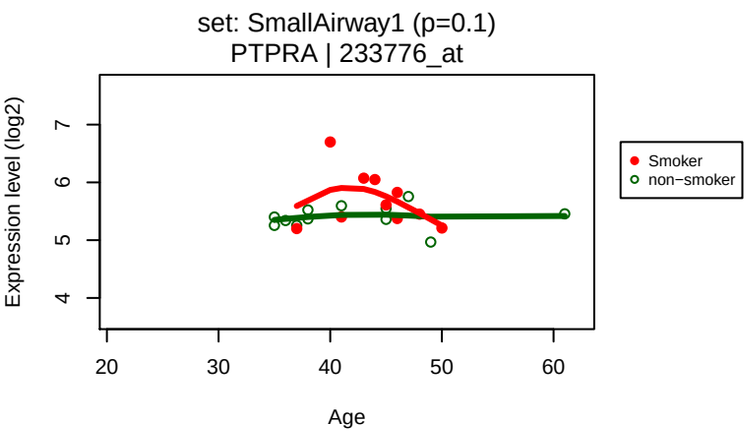
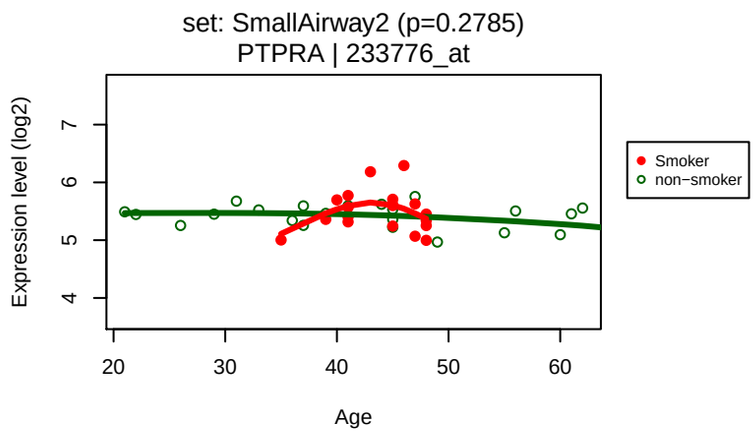
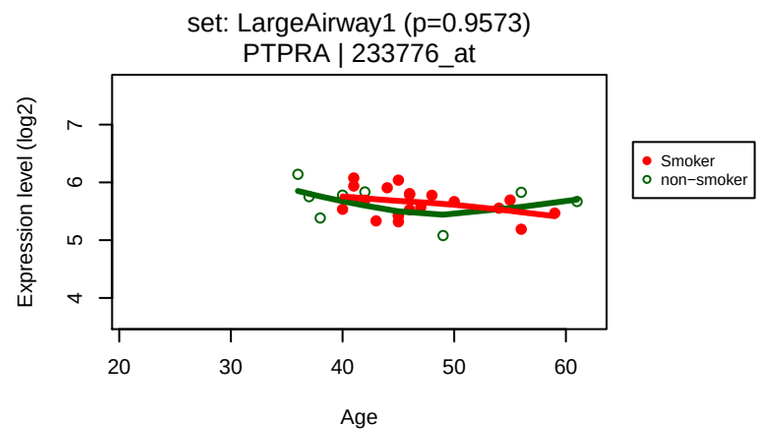

Gene expression of smokers and nonsmokers
gene: PTPRA | 233776_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1
p=0.2785
p=0.3896
p=0.9573
p=0.4622
p=0.0173
p=0.4699
p=0.4877
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

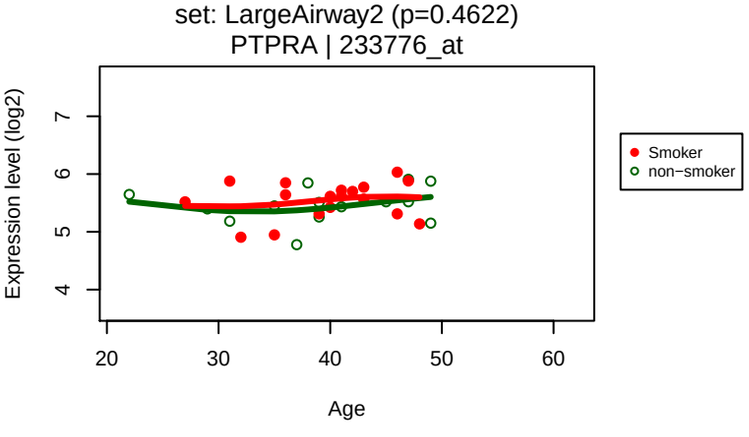

Gene expression ratio (smokers/nonsmokers)
gene: PTPRA | 233776_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1
p=0.2785
p=0.3896
p=0.9573
p=0.4622
p=0.0173
p=0.4699
p=0.4877
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

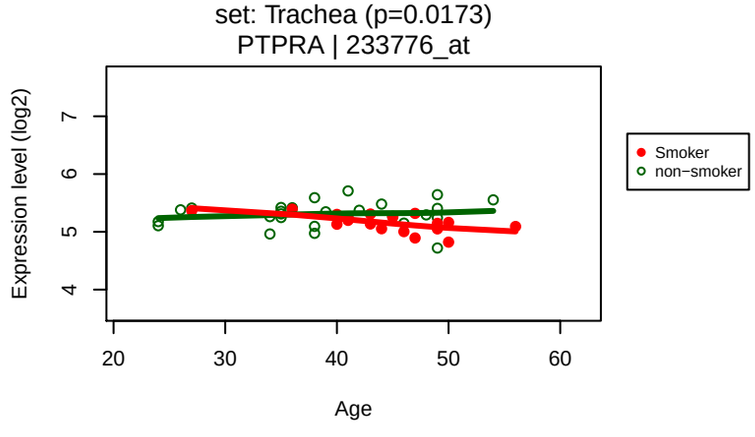

Gene expression ratio in Males
gene: PTPRA | 233776_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6585
p=0.3051
p=0.1462
p=0.4618
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTPRA | 233776_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4017
p=0.3813
p=0.188
p=0.8978
SmallAirway2
LargeAirway2
Trachea
Macrophage2

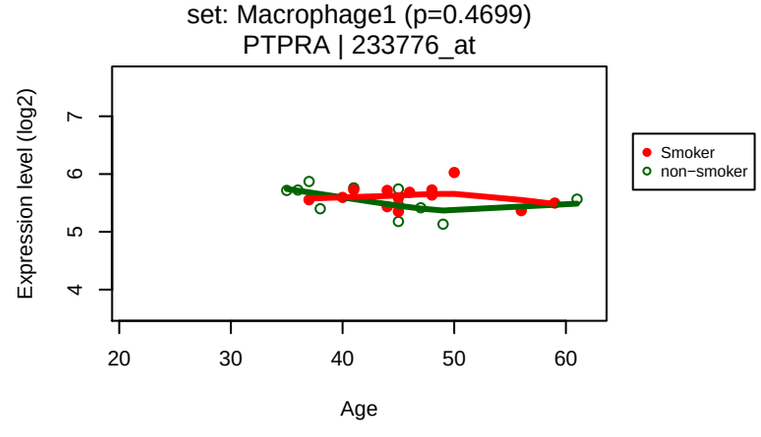

set: SmallAirway1 (p=0.1)
PTPRA | 233776_at
Expression level (log2)
Age
Smoker
non−smoker

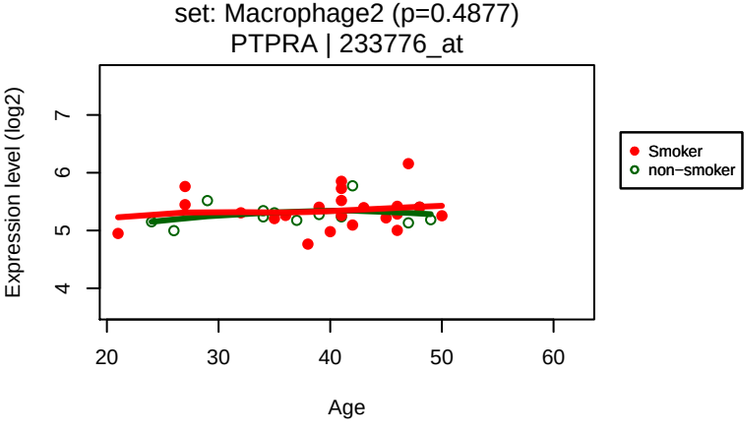

set: SmallAirway2 (p=0.2785)
PTPRA | 233776_at
Expression level (log2)
Age
Smoker
non−smoker

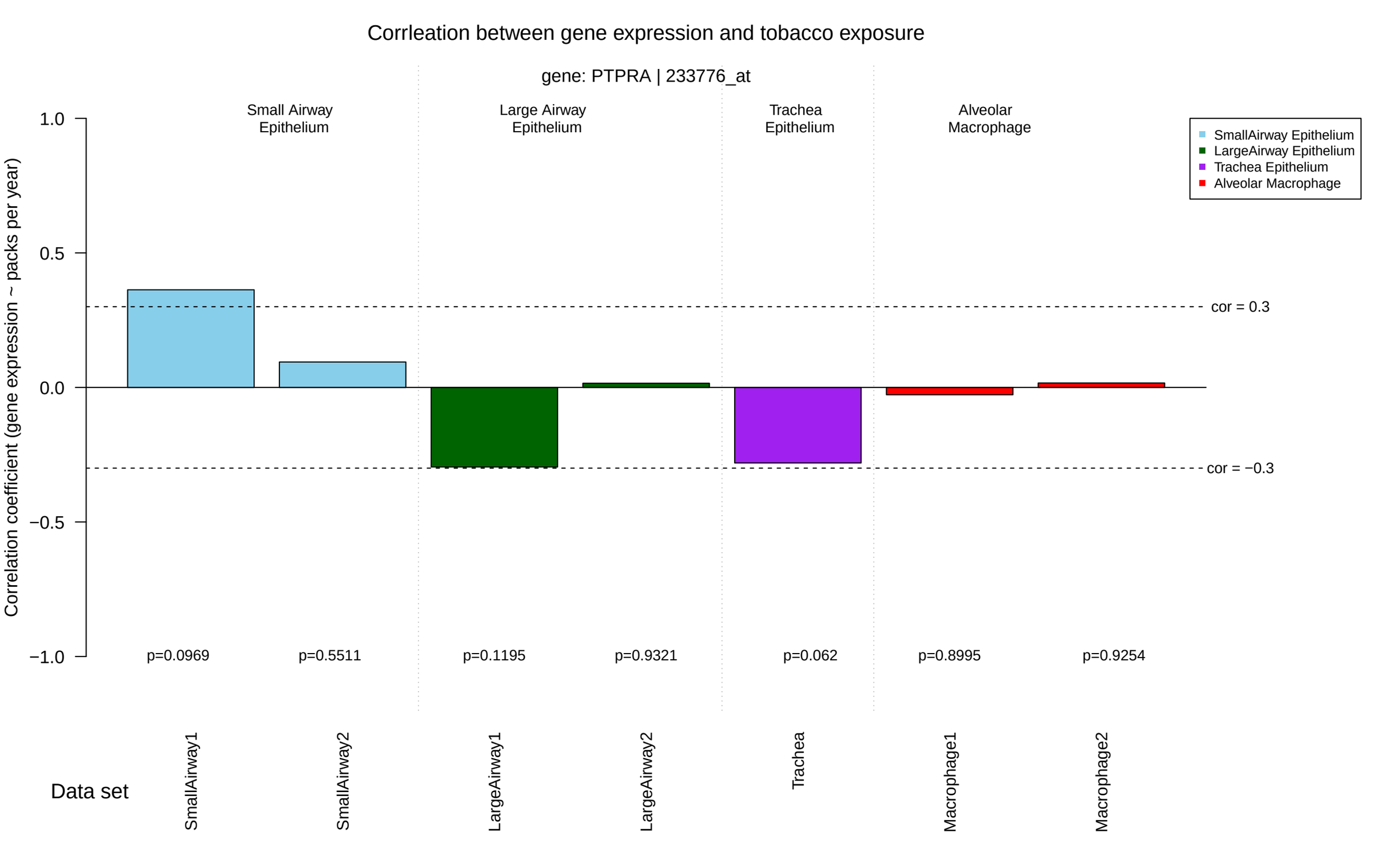

set: LargeAirway1 (p=0.9573)
PTPRA | 233776_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4622)
PTPRA | 233776_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0173)
PTPRA | 233776_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4699)
PTPRA | 233776_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4877)
PTPRA | 233776_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTPRA | 233776_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0969
p=0.5511
p=0.1195
p=0.9321
p=0.062
p=0.8995
p=0.9254
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2